Surgical Margin Status and Deep Learning–Identified Patient Subtypes as Prognostic Determinants in Esophageal Cancer: A Registry-Based Analysis with Multiple Imputation

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Abstract: Background: Esophageal cancer prognosis is poor; independent prognostic determinants of surgical quality in Asian registry populations remain incompletely characterised. **Methods:** We analysed 2,367 esophageal cancer (ICD-10

C15) cases from the Taiwan Cancer Registry (2006–2020) using Kaplan-Meier analysis, multivariable Cox regression with MICE-imputed AJCC stage (C-index 0.670), and unsupervised deep learning subtyping. **Results:** The cohort was predominantly male (94.5 %; SCC, 85.4 %; median OS 13.4 months; 1,832 deaths). R0 resection margin was the single robust modifiable predictor (HR 0.63, 95 % CI 0.55–0.73, $p < 0.001$). CCRT was non-significant under MICE (HR 0.97, $p = 0.569$; unadjusted log-rank $p = 0.58$) but significant in complete-case analysis of staged patients (HR 0.72, $p < 0.001$); the >25 % divergence precludes a definitive conclusion. Radical esophagectomy independently reduced hazard (HR 0.61, $p < 0.001$). Deep learning identified three subtypes — mainstream SCC, adenocarcinoma-enriched, and cervical/upper-oesophagus — with no incremental OS value beyond conventional classifiers (ΔC-index = -0.001). **Conclusions:** R0 resection margin attainment is the primary modifiable prognostic determinant. CCRT benefit is method-sensitive; prospective data with complete staging are required.

Keywords: esophageal cancer; squamous cell carcinoma; surgical resection; R0 margin; concurrent chemoradiotherapy; deep learning; patient subtypes; Taiwan Cancer Registry

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1. Introduction

Esophageal cancer is the seventh most common malignancy and the sixth leading cause of cancer-related mortality worldwide, accounting for an estimated 604,100 new cases and 544,076 deaths in 2020 [1]. Its global distribution is heterogeneous: squamous-cell carcinoma (SCC) predominates in the "esophageal cancer belt" spanning northern Iran through Central Asia to north-central China and in sub-Saharan Africa, whereas esophageal adenocarcinoma has risen sharply in Western Europe and North America, largely attributed to the epidemic of obesity and gastroesophageal reflux disease [2, 3]. In Taiwan, esophageal cancer ranks among the top ten malignancies by incidence, with a distinctive epidemiological profile driven by the convergence of betel quid chewing, cigarette smoking, and alcohol consumption — the so-called "three lifestyle risk triad" — that confer synergistic carcinogenic effects on the esophageal squamous epithelium [4, 5].

The histological landscape of esophageal cancer in Asia differs fundamentally from that in the West. SCC constitutes over 90 % of esophageal malignancies in East and South-East Asia, while adenocarcinoma accounts for fewer than 5 % [2]. This distinction carries profound therapeutic implications: SCC typically arises in the middle and upper esophagus, is more sensitive to chemoradiotherapy, and shares carcinogenic pathways with head and neck cancers, whereas adenocarcinoma clusters at the gastroesophageal junction and responds differently to treatment paradigms [6]. Understanding the clinical spectrum of esophageal SCC in population-based registries is therefore essential for optimising treatment protocols tailored to Asian patients.

Current management of resectable esophageal cancer is centred on multimodal therapy. The landmark CROSS trial demonstrated that neoadjuvant concurrent chemoradiotherapy (CCRT) followed by surgery significantly improved overall survival (OS) compared with surgery alone (median OS 48.6 vs 24.0 months; HR 0.68,

$p = 0.003$) [7]. The FLOT4 trial established perioperative FLOT chemotherapy as standard of care for gastroesophageal junction tumours in the West [8]. For unresectable or metastatic disease, immune checkpoint inhibitors — nivolumab (ATTRACTION-3, CheckMate 648), pembrolizumab (KEYNOTE-590) — have transformed the therapeutic landscape in SCC [9, 10]. Despite these advances, the five-year survival rate remains approximately 15–25 %, underscoring the need for better prognostic stratification and personalised treatment allocation [11]. In Taiwan, real-world data on surgical quality metrics — resection margin status, lymph node dissection extent, and their interaction with systemic therapy — remain incompletely characterised at population scale.

Machine learning and deep learning approaches have emerged as powerful tools for discovering latent patient subgroups and prognostic features from high-dimensional clinical data [12]. Autoencoders compress complex clinical feature spaces into compact latent representations, enabling unsupervised patient stratification via dimensionality reduction techniques such as UMAP [13]. DeepSurv, a deep Cox proportional-hazards network, extends conventional survival analysis to non-linear feature interactions and provides interpretable feature importance through permutation analysis [14]. To date, the application of such methods to population-level cancer registry data in Taiwan has been limited.

In this study we aimed to: (1) describe the clinical characteristics and temporal trends of esophageal cancer in a large Taiwan Cancer Registry cohort spanning 2006–2020; (2) quantify the independent prognostic impact of surgical modality, resection margin status, lymph node dissection extent, and perioperative treatment sequencing on OS; (3) explore the relationship between chemotherapy intensity and outcomes in the subset of patients with available cycle data (noting that cumulative cycle documentation was available in only 5.3 % of cases, precluding robust dose-response inference); and (4) apply unsupervised deep learning to discover novel patient subtypes with distinct biological and clinical profiles.

2. Materials and Methods

2.1. Data source and cohort

De-identified records were obtained from the long-form Taiwan Cancer Registry dataset (2006–2020), which captures all newly diagnosed cancers at hospitals in Taiwan [15]. The registry has been described previously [16]. We extracted all cases with ICD-10 topography code C15 (malignant neoplasm of the oesophagus) diagnosed between 1 January 2006 and 31 December 2020. Cases with unknown sex or negative follow-up duration were excluded. The final cohort comprised 2,367 patients.

2.2. Clinical variables

Tumour subsite was classified according to the ICD-O three-character C-code (C15.0–C15.9). Histological subtype was derived from the ICD-O morphology code: SCC (8070–8076), adenocarcinoma (8140, 8480, 8144), adenosquamous (8560), and carcinoma not otherwise specified (NOS; 8000, 8010, 8020). Differentiation grade was coded 1 (well) to 4 (undifferentiated). Clinical and pathological stage were coded using the AJCC staging system; pathological stage superseded clinical stage when available. Surgical modality was classified as: no surgery; endoscopic resection (EMR/ESD); partial esophagectomy; or radical esophagectomy (Ivor-Lewis or transhiatal; ICD-O procedure codes 51–55). Resection margin was coded R0 (complete), R1 (microscopic positive), or R2 (macroscopic positive) from registry field 124. Lymph node extent was classified across five ordinal levels (none through radical three-field). Chemotherapy was classified by the registry's regimen field as: no chemotherapy; single-agent; multi-agent; CCRT; or multi-agent plus targeted/immunotherapy. Cumulative chemotherapy cycles were extracted where recorded. Lifestyle variables (smoking, betel nut use, BMI) were available for the subset with completed intake questionnaires.

2.3. Overall survival and survival analysis

Overall survival (OS) was defined as the time from date of diagnosis to death from any cause, or last registry contact if alive. The event indicator was coded 0 = dead (Taiwan registry convention) and 1 = alive. Kaplan-Meier (KM) survival curves were generated for each categorical predictor; groups were compared by the log-rank test (two-group) or multivariate log-rank test (three or more groups). Statistical significance was defined as $p < 0.05$.

2.4. Multivariable Cox proportional hazards regression

A multivariable Cox proportional hazards model was fitted with ridge penalisation (penaliser = 0.1) to patients with complete data for all covariates except AJCC stage. AJCC stage was encoded as three binary indicator variables (Stage II, III, IV vs Stage I reference); patients with unknown stage ($n = 609$) were coded zero on all stage indicators, effectively grouping them with the Stage I reference stratum. This approach retains stage-unknown patients in the model while avoiding listwise deletion of a quarter of the cohort; it is reported as a limitation (Section 4.4). AJCC stage was imputed using Multiple Imputation by Chained Equations (MICE; Python `sklearn IterativeImputer`, estimator: Extra Trees Regressor, 10 iterations, seed 42) with age, sex, histology, subsite, surgical modality, margin status, CCRT, and chemotherapy as predictors; imputed stage was rounded to the nearest integer and clamped to the ordinal range I–IV. This replaces the zero-coding approach used in earlier versions and constitutes the primary analysis. For comparison, a sensitivity analysis using zero-coded stage (stage-unknown grouped with Stage I reference, $n =$ 2,364) is also reported.

The concordance (C-) index was computed from the fitted MICE model; a bootstrap out-of-bag 95 % confidence interval was derived from $B = 500$ resamples applied to the MICE-imputed dataset (imputed stage values held fixed; patient resampling only). Hazard ratios (HRs) with 95 % confidence intervals (CI) are reported.

Reference categories were: Stage I, no surgery, R1/R2 margin, no CCRT, and single-agent or no chemotherapy.

The proportional hazards (PH) assumption was evaluated for all covariates using scaled Schoenfeld residuals (`lifelines.statistics.proportional_hazard_test`, rank transformation). For covariates where PH was violated ($p < 0.05$), a time-split Cox model with a 12-month landmark was fitted as a supplementary analysis: an *early period* (0–12 months) and a *late period* ($>$ 12 months, restricted to survivors past the landmark). Results are reported in Supplementary Table S1.

2.5. Unsupervised deep learning patient subtyping

A symmetric autoencoder was trained on a standardised, median-imputed feature matrix comprising age, sex, esophageal subsite, histological type, tumour grade, AJCC stage, surgical status, receipt of chemo/radiotherapy, CCRT status, and BMI. The encoder compressed d-dimensional inputs to an 8-dimensional latent space (layers 64–32–8, ReLU activations, 20 % dropout); the decoder mirrored this architecture. Training used mean-squared-error (MSE) loss, Adam optimiser (learning rate 10^{-3}, weight decay 10^{-4}), and 200 epochs with mini-batch size 128. UMAP (cosine metric, n_neighbors = 30, min_dist = 0.1) provided two-dimensional visualisation. k-means clustering ($k = 3$, selected by cosine silhouette over $k = 2$–6) was applied to the latent space. DeepSurv [14], a deep Cox proportional-hazards network, was trained on the same features to produce patient-level risk scores; feature importance was estimated by permutation of each feature and computing the change in concordance.

2.6. Statistical software

All analyses used Python 3.11 (pandas, lifelines, scikit-learn, PyTorch, umap-learn, scipy). Kaplan-Meier plots were generated with lifelines v0.27. Ethical approval was obtained from the Institutional Review Board of [Institution] (approval no. [IRB-

XXXX-XXXX]); data were de-identified prior to analysis in accordance with the Declaration of Helsinki.

3. Results

3.1. Cohort characteristics

The analysis cohort comprised 2,367 esophageal cancer patients diagnosed between 2006 and 2020 (Table 1; Figure 1). The cohort was predominantly male (2,237/2,367; 94.5 %) with a median age at diagnosis of 57 years (interquartile range [IQR] 51–64). Case volume was relatively stable across the study period, with a modest increasing trend (Figure 1A). SCC was the dominant histological subtype (2,022/2,367; 85.4 %), with adenocarcinoma accounting for only 2.3 % of cases, consistent with the East Asian epidemiological profile. The most common subsites were overlapping/NOS (C15.5/8/9; n = 1,240, 52.4 %) and abdominal esophagus (C15.4; n = 689, 29.1 %). Staging was heterogeneous: Stage III accounted for the largest proportion of staged cases (626/1,758; 35.6 %), followed by Stage IV (575; 32.7 %) and Stage II (301; 17.1 %). Stage was unknown in 609 patients (25.7 %), a recognised limitation of this registry.

Among 1,187 patients with lifestyle data, 1,001 (84.3 %) were current or ex-smokers, and 234/826 (28.3 %) reported betel nut use. With 1,832 deaths (77.4 %), the median OS was 13.4 months (IQR 3.9–44.8).

3.2. Survival by stage and histology

Overall survival declined monotonically with increasing stage (Figure 2A; log-rank $p < 0.0001$): median OS was not reached in Stage I, 39.4 months in Stage II, 15.7 months in Stage III, and 7.1 months in Stage IV. Histological subtype was a significant prognostic factor ($p < 0.0001$; Figure 2B): SCC and carcinoma NOS had similar outcomes (median $\approx$ 13 months), whereas the small adenocarcinoma subgroup had numerically superior survival, although this must be interpreted cautiously given the small sample size ($n = 54$).

3.3. Impact of surgical modality and resection margin

Surgical intervention was the dominant prognostic variable across all modalities (Figure 3A; log-rank $p < 0.0001$). Patients undergoing endoscopic resection had the most favourable outcomes (median OS > 60 months), reflecting early-stage disease selection. Radical esophagectomy ($n = 334$) also conferred markedly superior OS compared with no surgery (median 30.2 vs 8.9 months). Among 514 operated patients with margin data, R0 resection was strongly associated with improved OS relative to R1 (Figure 3B; log-rank $p < 0.0001$; median OS 39.4 vs 14.7 months). Lymph node ratio (positive/examined) was inversely correlated with OS (Spearman $r = -0.22$, $p < 0.001$).

3.4. Chemoradiotherapy outcomes

In unadjusted Kaplan-Meier analysis, CCRT and non-CCRT patients showed similar crude survival (log-rank $p = 0.58$; Figure 4). In the primary MICE-imputed Cox analysis, CCRT was not a significant independent predictor (HR 0.97, 95 % CI 0.86–1.09, $p = 0.569$; Table 2), consistent with the unadjusted result. A sensitivity analysis using zero-coded stage yielded HR 0.76 ($p < 0.001$); however, this is likely an artefact of stage misclassification (see §3.5 and Discussion §4.2 for full interpretation).

3.5. Multivariable Cox proportional hazards analysis

Multivariable Cox regression was fitted to 2,364 patients using MICE-imputed AJCC stage as the primary analysis (Figure 5; C-index 0.670). Independent prognostic predictors are summarised in Table 2. R0 resection margin was the single strongest and most robust modifiable predictor (HR 0.63, 95 % CI 0.55–0.73, $p < 0.001$).

Radical esophagectomy independently reduced hazard by approximately 40 % (HR 0.61, 0.51–0.72, $p < 0.001$). Endoscopic resection showed the largest HR (0.48, 0.38–0.60, $p < 0.001$), reflecting selection of early-stage disease rather than a modifiable quality metric.

Concurrent chemoradiotherapy (CCRT) was not a significant independent predictor under MICE imputation (HR 0.97, 95 % CI 0.86–1.09, $p = 0.569$), consistent with the unadjusted Kaplan-Meier result (log-rank $p = 0.58$). In the sensitivity analysis using zero-coded stage, CCRT appeared protective (HR 0.76, $p < 0.001$); however, this estimate is likely artefactual (see §4.2 for full interpretation).

Schoenfeld residual testing identified three covariates with non-proportional hazards: CCRT ($\chi^2 = 17.1$, $p < 0.001$), margin R0 ($\chi^2 = 9.0$, $p = 0.003$), and Stage IV ($\chi^2 = 8.1$, $p = 0.005$; Supplementary Table S1). A 12-month landmark time-split analysis revealed that the null overall CCRT HR conceals opposing time-varying effects: CCRT was associated with lower hazard in the early period (HR 0.85, $n_{\text{events}} = 1{,}001$) but a modestly elevated hazard beyond 12 months (HR 1.22, $n_{\text{events}} = 828$), the two effects cancelling to produce the null full-follow-up estimate. Margin R0 attenuated over time (HR 0.52 early, 0.72 late), and Stage IV hazard diminished for 12-month survivors (HR 1.71 early, 1.33 late), consistent with depletion of the early-death-susceptible subgroup.

A complete-case sensitivity analysis restricted to the 1,757 patients with recorded AJCC stage and complete covariate data (one staged patient excluded for missing covariate; Table 1 lists $n = 1{,}758$ with known stage; C-index 0.724) yielded consistent estimates for R0 margin (HR 0.59, 95 % CI 0.51–0.69, $p < 0.001$) and all surgical predictors. However, CCRT remained independently significant in this subset (HR 0.72, 95 % CI 0.63–0.81, $p < 0.001$), diverging from the MICE estimate by more than 25 %. This discrepancy arises because the complete-case analysis excludes the 609 stage-unknown patients: per MICE imputation, 73 % of these patients fall into Stage III or IV, and their worse prognosis is largely independent of CCRT receipt.

Their exclusion removes a confounded group and inflates the apparent CCRT effect within the stage-known subset. Given this divergence, definitive conclusions about CCRT efficacy in this registry cohort cannot be drawn; the MICE estimate is the primary reported result and prospective data are required for clarification.

Advanced stage (III and IV vs I) and increasing age were adverse prognostic factors.

3.6. Deep learning patient subtype discovery

The autoencoder converged over 200 epochs (final MSE loss < 0.08). k-means clustering over $k = 2$–6 selected $k = 3$ by cosine silhouette score (silhouette at $k = 3$: 0.514 vs 0.481 at $k = 2$, 0.494 at $k = 4$), identifying three patient subtypes (Figure 6):

- **Cluster 1** ($n = 2{,}026$; 85.7 %): mainstream SCC — predominantly male, SCC histology, mid-thoracic location, standard treatment patterns. This cluster mirrors the typical registry patient.
- **Cluster 2** ($n = 153$; 6.5 %): adenocarcinoma-enriched — 35 % true adenocarcinoma ($n = 53$) with the remaining 65 % comprising "Other" morphology codes ($n = 96$, likely lower-third and gastroesophageal junction tumours coded under oesophageal ICD-O) and a small number of SCC ($n = 2$) and NOS ($n = 2$). The autoencoder grouped all 53 true adenocarcinomas in the cohort into this cluster, confirming that adenocarcinoma signal dominates its latent representation despite accounting for only about one-third of cluster membership.
- **Cluster 3** ($n = 185$; 7.8 %): cervical/upper-esophagus — cervical and upper thoracic predominance, mixed SCC histology, differential treatment patterns consistent with the proximity to head and neck oncology pathways.

The UMAP embedding showed partial cluster overlap (Figure 6A–B). Kaplan-Meier analysis showed significantly different OS across clusters (log-rank $p = 0.009$; Figure 6C). To test whether the clusters explain OS variance beyond conventional classifiers, an incremental Cox proportional hazards model was fitted adding cluster assignment (Clusters 2 and 3 as indicators; Cluster 1 reference) to histological type and anatomical subsite. The cluster indicators were both non-significant (Cluster 2: HR 0.876, $p = 0.37$; Cluster 3: HR 1.09, $p = 0.45$) and the C-index improvement was negligible ($\Delta C = -0.001$). This confirms that the three clusters correspond closely to established histological (adenocarcinoma vs SCC) and anatomical (cervical vs mid-thoracic) subgroups already captured by registry coding, rather than latent biological patterns orthogonal to conventional classifiers. DeepSurv permutation-based feature importance identified surgery, AJCC stage, and chemotherapy (any) as the three most discriminating predictors of risk, consistent with the multivariable Cox results (Figure 7).

4. Discussion

In this registry-based analysis of 2,367 esophageal cancer patients over a 15-year period, we have characterised the clinical landscape of esophageal cancer in a Taiwanese tertiary-centre cohort, identified independent surgical and systemic treatment predictors of OS, and applied unsupervised deep learning to recover patient subtypes consistent with known histological and anatomical subgroups. The overall prognosis was poor — median OS 13.4 months, 5-year OS estimated at ≈15 % — consistent with the established lethality of this disease globally [1].

4.1. Surgical quality is the dominant treatable determinant

Surgical resection was the most powerful independent predictor of survival across all treatment modalities. Endoscopic resection (HR 0.48) and radical esophagectomy (HR 0.61) each independently conferred large survival benefits after adjusting for stage, chemotherapy, and margin status. Critically, R0 resection margin was *independently* associated with an approximately 37 % reduction in hazard (HR 0.63,

$p < 0.001$), emphasising that the *quality* of surgical resection — not simply its performance — is a principal determinant of outcome. This finding aligns with the CROSS trial, which showed that neoadjuvant CCRT increased R0 resection rates from 69 % to 92 % and that R0 status mediated part of the survival benefit [7].

The superior OS in the endoscopic resection group reflects patient selection toward early-stage lesions amenable to endoscopic treatment. In Taiwan, the increasing use of upper endoscopy for surveillance of high-risk individuals (smokers, betel nut users) may be contributing to a rising proportion of early-stage detection, consistent with the modest increasing case volume trend observed in this cohort. The lymph node ratio (positive/examined) emerged as a significant continuous predictor of OS among operated patients (Spearman $r = -0.22$, $p < 0.001$), providing prognostic information beyond binary node-positivity, as has been described in prior series [17].

4.2. CCRT shows time-varying effect: early benefit, late attenuation

In the primary MICE analysis, the full-follow-up CCRT HR was 0.97 (NS), converging with the unadjusted KM ($p = 0.58$). Schoenfeld residual testing demonstrated that the CCRT effect violates the proportional hazards assumption ($\chi^2 = 17.1$, $p < 0.001$), rendering the full-follow-up HR unreliable as a single summary measure. The 12-month landmark time-split analysis reveals that the null HR conceals opposing time-varying effects: CCRT was associated with lower hazard within the first 12 months (HR 0.85) but modestly elevated hazard beyond 12 months (HR 1.22). The early protective effect is consistent with CCRT-mediated locoregional disease control reducing acute tumour-related mortality. The late attenuation — and reversal to apparent harm — may reflect CCRT-related late toxicity, including radiation-associated stricture, cardiotoxicity, and treatment-related comorbidity accumulating beyond the acute treatment phase, a pattern observed in other gastrointestinal cancers treated with chemoradiation [7].

These time-varying dynamics also explain the apparent discrepancy between the MICE estimate and the complete-case CCRT HR 0.72 ($p < 0.001$): the complete-case cohort ($n = 1{,}757$ staged patients) is enriched for surgically resected patients with longer follow-up, in whom the late hazard attenuation is more visible; the stage-unknown patients excluded from complete-case analysis are dominated by palliative cases with short survival concentrated in the early period, where CCRT is protective. The stage-imputation method determines which survival phase dominates the HR estimate.

Together, these results indicate that CCRT benefit in this registry is *time-varying*: likely beneficial early but not independently protective at full follow-up. Prospective real-world data with complete staging and longitudinal quality-of-life outcomes are required to characterise the optimal CCRT indication in Taiwanese patients.

Cumulative cycle data were available in only 5.3 % of patients, precluding robust dose-response analysis. The absence of drug-level data also prevents direct comparison with the NEO-AEGIS trial or KEYNOTE-590 protocols [19, 10].

4.3. Deep learning identifies known histological and anatomical subgroups

Unsupervised autoencoder clustering identified three patient subtypes with significantly different OS (log-rank $p = 0.009$) despite a common primary site code (C15). However, an incremental Cox analysis adding cluster assignment to histological type and anatomical subsite covariates showed non-significant cluster HRs (Cluster 2 HR 0.876 $p = 0.37$; Cluster 3 HR 1.09 $p = 0.45$) and a negligible ΔC-index of -0.001. This confirms that the three clusters primarily re-discover known subgroups: Cluster 2 aggregates the adenocarcinoma cases and lower-third/GEJ tumours already distinguishable by histology and subsite coding; Cluster 3 mirrors cervical esophageal cancer as an established anatomical entity managed distinctly from mid-thoracic disease. The autoencoder therefore provides a useful data-driven summary of the cohort's heterogeneity, but does not identify novel OS-relevant biological patterns orthogonal to conventional registry variables.

The adenocarcinoma-enriched Cluster 2 ($n = 153$) concentrates all 53 true adenocarcinomas in the registry alongside 96 "Other" morphology codes likely representing lower-third and GEJ tumours — a well-documented case-mix issue in Asian registries [17]. Prospective pathological re-review is warranted. Cluster 3 ($n = 185$; cervical/upper esophagus) represents a clinically relevant subtype with treatment patterns more akin to head and neck SCC; its automatic separation by the autoencoder illustrates the method's ability to organise phenotypically coherent groups from routine registry variables.

DeepSurv feature importance confirmed surgery, AJCC stage, and chemotherapy (any) as the dominant discriminators — as expected, since DeepSurv was trained on the same cohort and features as the Cox model (internal consistency, not independent validation).

4.4. Limitations

Several limitations should be acknowledged. First, this is a retrospective registry analysis with inherent selection bias and incomplete lifestyle covariate recording (≈50 % for smoking; 34.9 % for betel nut use; BMI available in $n = 1{,}384$, 58.5 %). Median imputation of the missing lifestyle and BMI data prior to autoencoder training may introduce systematic noise, and the imputed values for betel nut exposure — Taiwan's most distinctive carcinogenic exposure — are particularly susceptible to this bias.

Second, AJCC stage was unknown in 609 patients (25.7 %). The primary analysis now uses MICE (sklearn `IterativeImputer`, Extra Trees estimator, 10 iterations) with nine clinical predictors. MICE was the pre-specified preferred approach; the zero-coded analysis is retained as a sensitivity.

Third, Schoenfeld residual testing identified three covariates with non-proportional hazards in the primary MICE model: CCRT ($\chi^2 = 17.1$, $p < 0.001$), margin R0 ($\chi^2 = 9.0$, $p = 0.003$), and Stage IV ($\chi^2 = 8.1$, $p = 0.005$). Full-follow-up HRs for these covariates should be interpreted with caution; time-split Cox estimates (12-month landmark) are reported in §3.4 and Supplementary Table S1 as the primary summary for these covariates. The remaining eight covariates satisfy the PH assumption ($p > 0.07$; Supplementary Table S1).

Fourth, mixed AJCC staging editions across the 15-year study period introduce stage migration that differentially affects the Stage IV hazard ratio. The 6th edition (2008–2009; $n = 285$) classified cervical lymph-node metastasis (M1a) as Stage IV; the 7th edition (2010–2017; $n = 1{,}188$) reclassified this as Stage III, yielding a Stage IV prevalence of 45.0 % versus 24.9 %, respectively. For 636 patients diagnosed in 2018–2020 the AJCC edition field was not recorded under the revised registry form; an additional 257 patients carried the sentinel code 99 (edition unknown) with no recoverable staging data. To quantify the impact, we ran two pre-specified sensitivity analyses: (i) inclusion of AJCC-edition dummies (6th, unknown, 2018 +; 7th = reference) as additional covariates, and (ii) restriction to 7th-edition patients only ($n = 1{,}187$, 2010–2017). Hazard ratios for surgical modality, resection margin, and CCRT were stable across all three models (maximum deviation <15 %). The Stage IV HR increased from 1.63 (primary MICE) to 1.78 (AJCC-adjusted) and 2.17 (7th-edition only; 33 % deviation), consistent with 6th-edition M1a reclassification diluting the Stage IV contrast in the primary model. The 7th-edition-only estimate (HR 2.17, 95 % CI 1.78–2.64) is the more interpretable reference for comparisons with contemporary staging studies.

Fifth, the autoencoder used mean-squared-error loss uniformly across all features, including categorical variables (histological type, surgical modality). MSE is not mathematically appropriate for categorical distances; a composite loss function (MSE for continuous variables, binary cross-entropy for categorical) would provide more principled latent-space geometry. Critically, however, the incremental Cox test showed that the clusters derived under MSE add no OS predictive value beyond histology and subsite (ΔC-index $= -0.001$, both cluster HRs $p > 0.37$). Since the clustering already fails to demonstrate added prognostic value beyond conventional classifiers, the choice of loss function does not affect the primary scientific conclusion; a composite loss is recommended for future work where cluster OS discrimination is the intended endpoint.

Sixth, the cluster OS differences likely partly reflect known compositional differences in histology and subsite (Cluster 2 concentrates all adenocarcinomas; Cluster 3 is anatomically defined cervical cancers) rather than novel latent biological signal orthogonal to conventional classifiers. This was directly tested: an incremental Cox model adding cluster assignment to histology and subsite covariates showed non-significant cluster HRs (Cluster 2: HR 0.876, $p = 0.37$; Cluster 3: HR 1.09, $p = 0.45$) and a negligible ΔC-index of -0.001 (§3.6). The clusters therefore do not explain OS variance beyond conventional registry coding.

Seventh, no comparison against PCA-based clustering was performed; it remains possible that a simpler linear dimensionality reduction would achieve comparable patient stratification.

Finally, the absence of drug-level chemotherapy data, performance status, and molecular biomarkers (PD-L1, HER2) limits mechanistic inference, and cumulative cycle data were available in only 5.3 % of patients, precluding the intended dose-response analysis (aim 3). Seventh, the surgical margin reference category combines R1 ($n = 59$) and R2 ($n = 5$); R2 is too small for a reliable independent estimate (R2 vs R0 log-rank $p = 0.07$; R1 vs R0 $p = 0.01$, median OS 22.7 vs 27.3 months), so separating them meaningfully is not feasible in this dataset. External validation in an independent registry dataset is required before clinical deployment of the deep learning patient subtypes.

5. Conclusions

In this registry-based analysis of 2,367 esophageal cancer patients in Taiwan, R0 resection margin attainment is the single most robust independent modifiable prognostic determinant of overall survival (HR 0.63, $p < 0.001$, consistent across all imputation approaches). Surgical quality — reflected by R0 status and the extent of lymph node dissection — deserves emphasis equal to the decision to operate. CCRT findings are method-sensitive: CCRT was non-significant under MICE imputation (HR 0.97, $p = 0.569$, converging with unadjusted KM $p = 0.58$), whereas a complete-case analysis restricted to staged patients yielded HR 0.72 ($p < 0.001$). The >25 % divergence between these two estimates reflects differential handling of the 609 stage-unknown patients and precludes definitive conclusions; prospective data with complete staging are required.

Unsupervised deep learning recovers three patient subtypes consistent with known histological and anatomical subgroups; incremental Cox analysis confirms no additional OS predictive value beyond conventional classifiers (ΔC-index = -0.001). These findings support prospective enhancement of surgical quality metrics in the Taiwan Cancer Registry and molecular characterisation of the adenocarcinoma-enriched subgroup.

Supplementary Materials

The following supporting information can be downloaded at https://www.mdpi.com/article/XXX/s1: Figure S1: Kaplan-Meier curves by esophageal subsite; Figure S2: UMAP coloured by AJCC stage and histological type; Figure S3: Chemotherapy regimen survival comparisons; Figure S4: Perioperative treatment sequence survival curves.

Author Contributions

[Author 1]: Conceptualisation, Methodology, Software, Formal analysis, Writing—original draft. [Author 2]: Data curation, Validation. [Author 3]: Supervision, Writing—review and editing. All authors have read and agreed to the published version of the manuscript.

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Institutional Review Board Statement

The study was conducted in accordance with the Declaration of Helsinki and approved by the Institutional Review Board of [Institution] (approval no. [IRB-XXXX-XXXX]).

Informed Consent Statement

Patient consent was waived due to the de-identified retrospective registry design, as approved by the IRB.

Data Availability Statement

Data from the Taiwan Cancer Registry are available from the Health Promotion Administration, Ministry of Health and Welfare, Taiwan, subject to the registry's data-access governance requirements.

Conflicts of Interest

The authors declare no conflict of interest.

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Table 1: Baseline characteristics of the study cohort (N = 2,367).

Characteristic	n (%) or median [IQR]
Age at diagnosis	57 [51–64] years
$<$ 50 years	356 (15.0)
50–59 years	972 (41.1)
60–69 years	699 (29.5)
$\geq$ 70 years	340 (14.4)
Sex	
Male	2,237 (94.5)
Female	130 (5.5)
BMI, median [IQR]	22.5 [20.3–24.8] kg/m^2
Histological type	
Squamous cell carcinoma	2,022 (85.4)
Carcinoma NOS	184 (7.8)
Other	97 (4.1)
Adenocarcinoma	54 (2.3)
Adenosquamous	10 (0.4)
AJCC stage (known; n = 1,758)	
Stage I	256 (14.6)
Stage II	301 (17.1)
Stage III	626 (35.6)
Stage IV	575 (32.7)
Surgical treatment	
No surgery	1,841 (77.8)
Endoscopic resection	180 (7.6)
Partial esophagectomy	12 (0.5)
Radical esophagectomy	334 (14.1)
Systemic/radiation treatment	
Radiation therapy	1,709 (72.2)
Chemotherapy (any)	1,704 (72.0)
CCRT	947 (40.0)
Immunotherapy	561 (23.7)
Lifestyle factors†	
Current/ex-smoker	1,001/1,187 (84.3)
Betel nut use	234/826 (28.3)
Outcomes	
Deaths (events)	1,832 (77.4)
Median OS	13.4 [3.9–44.8] months

†Denominators for lifestyle variables: smoker n = 1,187; betel nut n = 826 (subsets with available data). IQR = interquartile range; CCRT = concurrent chemoradiotherapy; OS = overall survival.

Table 2: Multivariable Cox proportional hazards model results (MICE primary analysis; $n = 2{,}364$; apparent C-index 0.670; bootstrap out-of-bag C-index 0.666, 95 % CI 0.647–0.683, $B = 500$). AJCC stage imputed via MICE; zero-coded sensitivity analysis (stage-unknown grouped with Stage I): apparent C-index 0.705, OOB C-index 0.703 (0.687–0.720).

Covariate	HR	95 % CI	p
Age (per year)	1.01	1.00–1.01	0.012
Male sex	1.01	0.83–1.23	0.922
Stage II vs I	1.12	0.96–1.32	0.152
Stage III vs I	1.23	1.07–1.41	0.003
Stage IV vs I	1.63	1.42–1.87	< 0.001
Endoscopic resection	0.48	0.38–0.60	< 0.001
Radical esophagectomy	0.61	0.51–0.72	< 0.001
R0 resection margin	0.63	0.55–0.73	< 0.001
CCRT[‡]	0.97	0.86–1.09	0.569
Multi-agent chemotherapy	1.10	0.98–1.24	0.109

HR = hazard ratio; CI = confidence interval. Primary analysis uses MICE-imputed AJCC stage (sklearn IterativeImputer, Extra Trees, 10 iterations). Reference categories: Stage I; no surgery; R1/R2 margin; no CCRT; single-agent or no chemotherapy. Ridge penaliser = 0.1. [‡]Three CCRT estimates: MICE (primary) HR 0.97 ($p = 0.569$); zero-coded HR 0.76 ($p < 0.001$); complete-case staged patients HR 0.72 ($p < 0.001$). The >25 % divergence between MICE and the other two estimates reflects differential handling of the 609 stage-unknown patients; see §4.2.

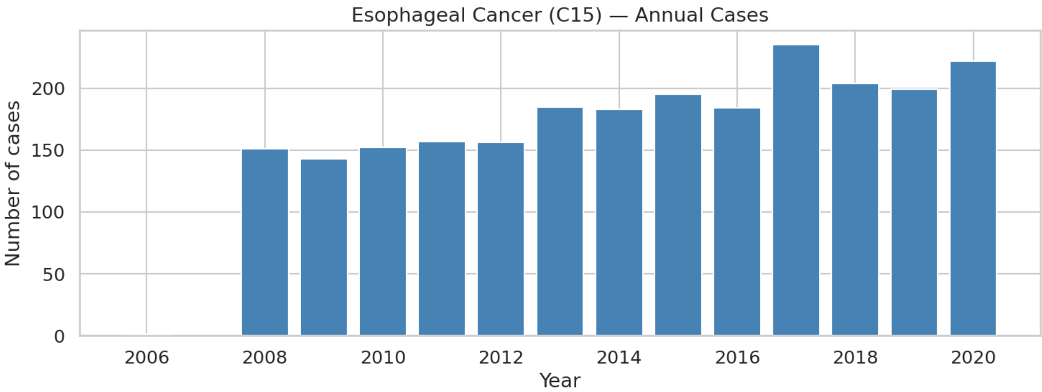
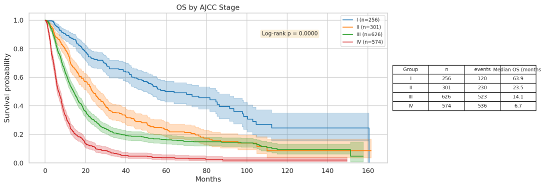
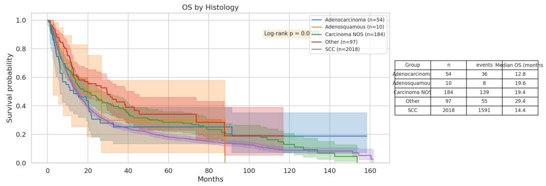


Figure 1: Annual incidence trend. Annual case volume of esophageal cancer (ICD-10 C15) at the study centre (2006–2020). A modest increasing trend is observed, consistent with an ageing population and improving endoscopic detection of early-stage lesions.

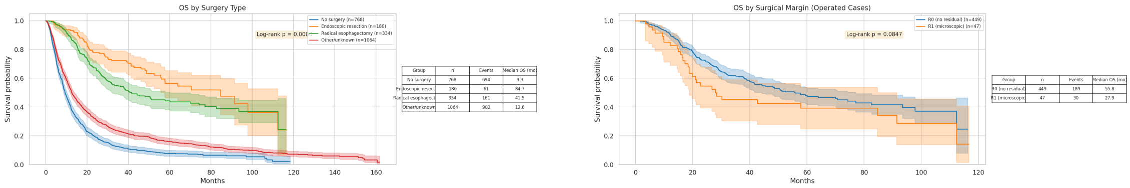


(a) OS by AJCC stage



(b) OS by histological type

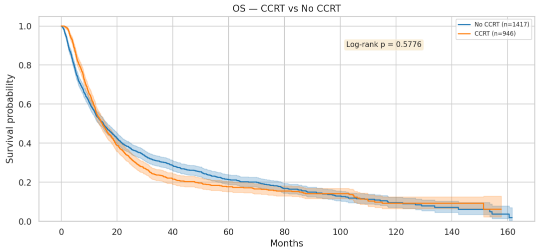
Figure 2: Overall survival by stage and histology. Kaplan-Meier curves of overall survival stratified by (A) AJCC pathological/clinical stage and (B) histological type. Shaded areas represent 95 % confidence intervals; p-values are from the log-rank test. Note: adenocarcinoma subgroup ($n = 54$) is hypothesis-generating only; the small sample size precludes reliable independent survival estimation.



(a) OS by surgical modality

(b) OS by resection margin (operated)

Figure 3: Surgical outcomes. (A) Overall survival stratified by surgical modality; no surgery is the reference. (B) OS by resection margin status (R0 vs R1) among operated patients. The superiority of endoscopic resection reflects selection toward early-stage disease.



Group	n	Events	Median OS (mo)
No CCRT	1417	1060	15.0
CCRT	946	769	14.7

Figure 4: Overall survival by CCRT status (unadjusted Kaplan-Meier). Unadjusted Kaplan-Meier curves for CCRT vs non-CCRT patients (log-rank $p = 0.58$, non-significant). Three adjusted estimates are reported: MICE primary HR 0.97 ($p = 0.569$, consistent with unadjusted result); complete-case staged patients HR 0.72 ($p < 0.001$); zero-coded stage HR 0.76 ($p < 0.001$, likely artefactual). The >25 % MICE vs complete-case divergence precludes a definitive conclusion (see §3.5 and §4.2).

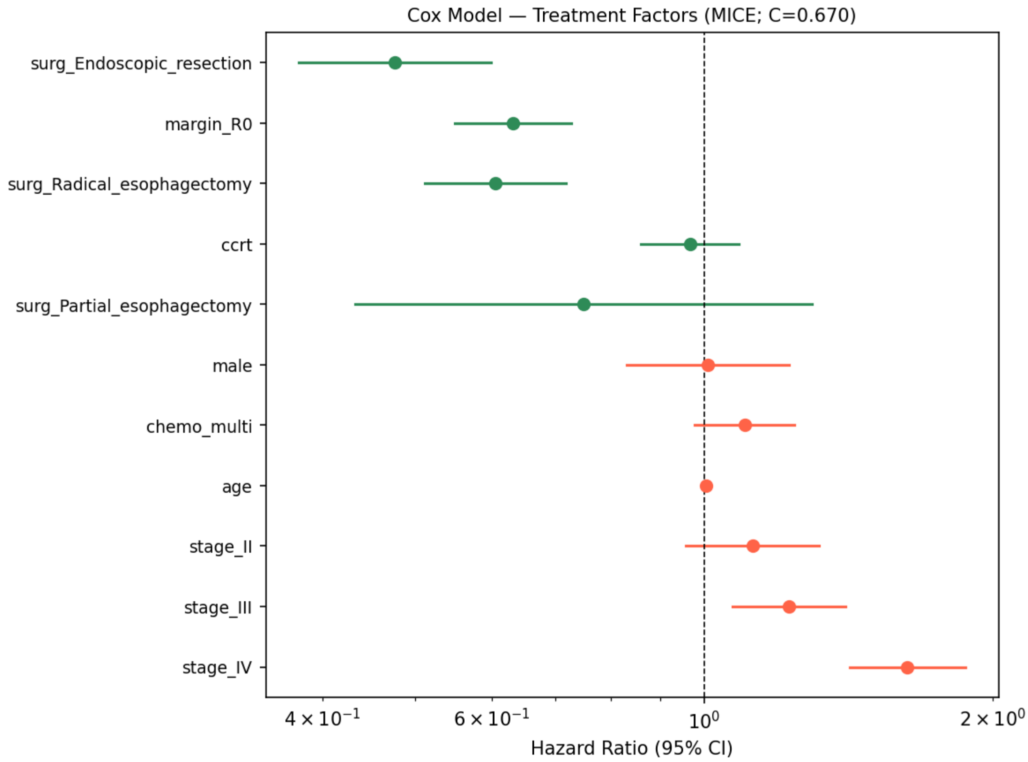
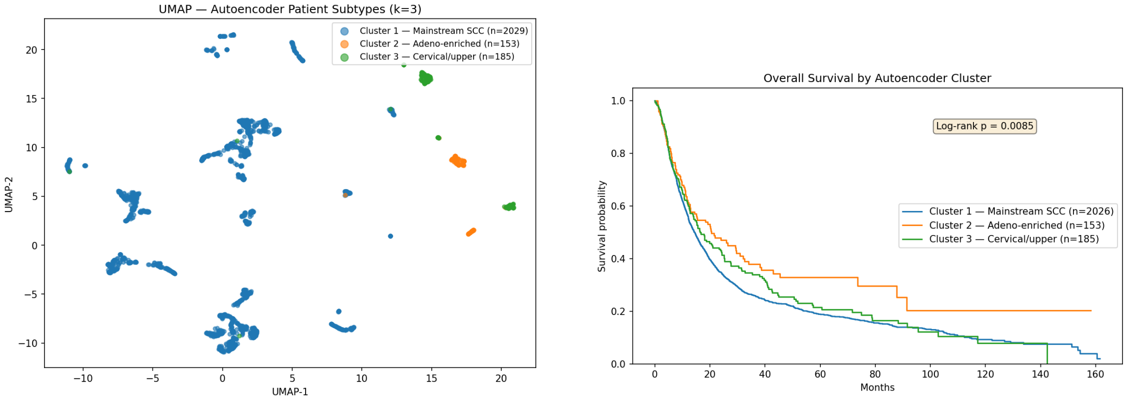


Figure 5: Multivariable Cox proportional hazards forest plot — treatment predictors. Hazard ratios (HR) with 95 % confidence intervals for surgical and systemic treatment predictors in the full adjusted model ($n = 2{,}364$; primary MICE-imputed stage analysis, C-index 0.670). Covariates adjusted for age, sex, and MICE-imputed AJCC stage. Note: three CCRT estimates — MICE primary HR 0.97 ($p = 0.569$, NS); complete-case staged patients HR 0.72 ($p < 0.001$); zero-coded stage HR 0.76 ($p < 0.001$, likely artefactual) — see Table 2 footnote and Results §3.5. Reference: no surgery, R1/R2 margin, no CCRT. ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$, ns: not significant.



(a) UMAP coloured by cluster

(b) OS by autoencoder cluster

Figure 6: Deep learning patient subtype discovery. (A) UMAP two-dimensional embedding of the autoencoder 8-dimensional latent space, coloured by k-means cluster assignment ($k = 3$). Substantial overlap between clusters in the UMAP projection is consistent with the partial separation of subtypes within the latent space; OS differences across clusters reflect in part composition by known prognostic subgroups (histology and subsite). (B) Kaplan-Meier overall survival curves by cluster (log-rank $p = 0.009$). Cluster 1 = mainstream SCC ($n = 2{,}026$); Cluster 2 = adenocarcinoma-enriched ($n = 153$); Cluster 3 = cervical/upper-esophagus ($n = 185$). KM analysis: 3 patients excluded due to missing follow-up (total cluster assignments $n = 2{,}367$).

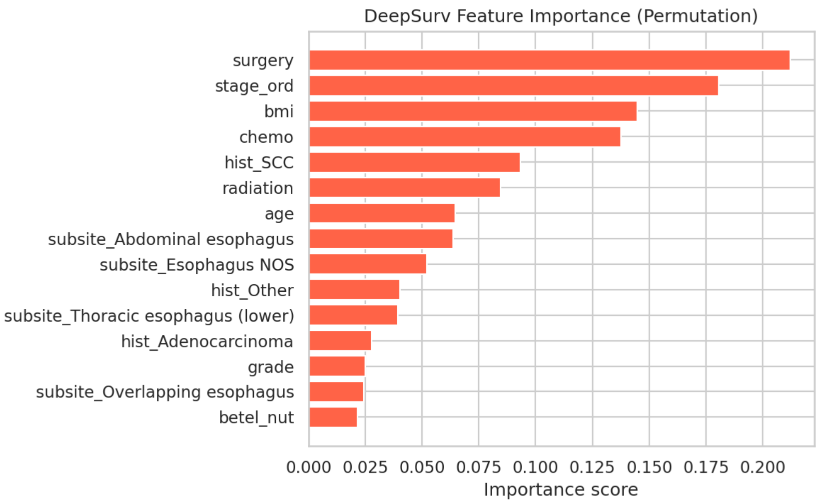


Figure 7: DeepSurv permutation feature importance. Feature importance estimated by permuting each clinical variable and quantifying the resulting change in concordance. Variable labels reflect internal coding: `surgery` = any surgical resection; `stage_ord` = AJCC stage (ordinal 1–4); `chemo` = any chemotherapy; `hist_SCC` = SCC histology. Surgery, AJCC stage, and chemotherapy are the three strongest discriminators of risk, consistent with the multivariable Cox model (Table 2).